$L = 2\ell$ periodic functions

Let $f(x) = f(x+L)$, $L = 2\ell$. E.g. the trigonometric functions

$$\cos\frac{n\pi x}{\ell}, \qquad \text{and} \qquad \sin\frac{n\pi x}{\ell}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{\ell} + b_n \sin\frac{n\pi x}{\ell}\right)$$

$$\int_{-\ell}^{\ell} \cos\frac{n\pi x}{\ell}\cos\frac{m\pi x}{\ell}\,\mathrm{d}x = \ell\delta_{nm} \qquad \int_{-\ell}^{\ell} \sin\frac{n\pi x}{\ell}\sin\frac{m\pi x}{\ell}\,\mathrm{d}x = \ell\delta_{nm}, \quad \{n, m\} \in \mathbb{N}$$

$$\int_{-\ell}^{\ell} \cos\frac{n\pi x}{\ell}\sin\frac{m\pi x}{\ell}\,\mathrm{d}x = \int_{-\ell}^{\ell} \cos\frac{n\pi x}{\ell}\,\mathrm{d}x = \int_{-\ell}^{\ell} \sin\frac{n\pi x}{\ell}\,\mathrm{d}x = 0, \quad \forall n \in \mathbb{N}$$

$$a_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(x)\cos\frac{n\pi x}{\ell}\,\mathrm{d}x \qquad n \in \{0, 1, 2, \dots\}$$

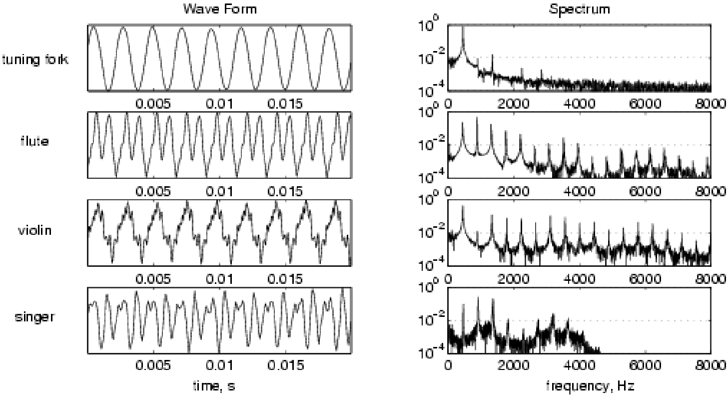
$$b_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(x)\sin\frac{n\pi x}{\ell}\,\mathrm{d}x \qquad n \in \{1, 2, \dots\}$$

For symmetric $f(x) = f(-x)$ functions $b_1 = b_2 = \cdots = 0$ and

$$\boxed{a_n = \frac{2}{\ell}\int_0^\ell f(x)\cos\frac{n\pi x}{\ell}\,\mathrm{d}x \qquad n \in \{0, 1, 2, \dots\}}.$$

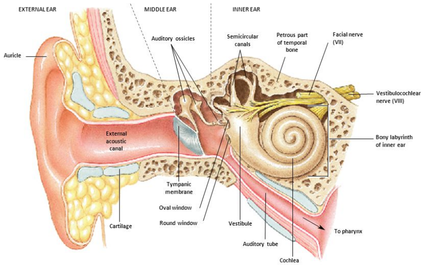
For antisymmetric $f(x) = -f(-x)$ functions $a_0 = a_1 = a_2 = \cdots = 0$ and

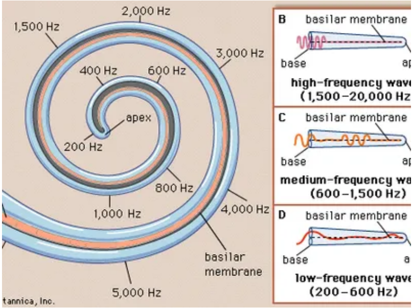
$$\boxed{b_n = \frac{2}{\ell}\int_0^\ell f(x)\sin\frac{n\pi x}{\ell}\,\mathrm{d}x \qquad n \in \{1, 2, \dots\}}.$$



- fundamental/base frequency → pitch
- harmonics → timbre/quality

Anatomy of the Ear





Complex Fourier series

$$\left.\begin{aligned} e^{ix} &= \cos x + i\sin x \\ \overline{e^{ix}} = e^{-ix} &= \cos x - i\sin x \end{aligned}\right\} \Rightarrow \begin{cases} \cos x = \dfrac{e^{ix}+e^{-ix}}{2} \\ \sin x = \dfrac{e^{ix}-e^{-ix}}{2i} \end{cases}$$

$f : [-\pi, \pi] \longrightarrow \mathbb{R}$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \frac{e^{inx}+e^{-inx}}{2} + b_n \frac{e^{inx}-e^{-inx}}{2i} \right)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2} e^{inx} + \frac{a_n + ib_n}{2} e^{-inx} \right).$$

Let us introduce the following coefficients : $c_n = \dfrac{a_n - ib_n}{2}$, $\overline{c}_n = c_{-n} = \dfrac{a_n + ib_n}{2}$ and $c_0 = \dfrac{a_0}{2}$.

Orthogonality of e^{inx}

$$\int_{-\pi}^{\pi} e^{inx}\overline{e^{imx}}\,\mathrm{d}x \equiv \int_{-\pi}^{\pi} e^{i(n-m)x}\,\mathrm{d}x = 2\pi\delta_{nm}.$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n\, e^{inx}$$

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y) e^{-iny}\,\mathrm{d}y$$

$L = 2\ell$ periodic functions

If $2\pi \to 2\ell$, $nx \to \dfrac{n\pi x}{\ell}$

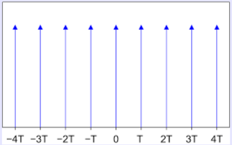
$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{n\pi x}{\ell}}.$$

The norm of these basis functions is 2ℓ

$$c_n = \frac{1}{2\ell}\int_{-\ell}^{\ell} f(y)e^{-i\frac{n\pi y}{\ell}}\,dy$$

Definition: Dirac comb

$$Ш_T(t) \equiv T \cdot \sum_{n=-\infty}^{\infty} \delta(x - nT)$$



Note that $Ш_T$ is dimensionless, irrespective of x's dimension.

Example: Dirac comb

For the Dirac comb $Ш_{2\pi}(x)$ the complex Fourier coefficients are

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} 2\pi\,\delta(x) e^{-inx} \mathrm{d}x = 1 \ .$$

Substituting back to the Fourier series we get

$$\boxed{\sum_{n=-\infty}^{\infty} e^{inx} = Ш_{2\pi}(x)}$$

More generally

$$\boxed{\sum_{n=-\infty}^{\infty} e^{in\frac{2\pi}{L}x} = L \cdot \sum_{n=-\infty}^{\infty} \delta(x - nL)}$$

Fourier-transform

$$f : (-\ell,\ \ell) \longrightarrow \mathbb{R}\ , \qquad 2\ell \text{ periodic function}$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{-i\frac{n\pi x}{\ell}}\ .$$

$$c_n = \frac{1}{2\ell}\int_{-\ell}^{\ell} f(y) e^{i\frac{n\pi y}{\ell}}\ dx$$

Non-periodic function = infinite periodic function, $\ell \to \infty$

Introducing the notation $k_n \equiv \frac{n\pi}{\ell}$ and using $\Delta k = k_{n+1} - k_n = \frac{\pi}{\ell}$:

$$f(x) = \frac{1}{2\pi}\sum_{k_n}\left[\int_{-\ell}^{\ell} f(y) e^{ik_n y}\ dy\right] e^{-ik_n x}\Delta k$$

$$\lim_{\ell\to\infty}\sum_{k_n} \ldots \text{function}(k_n)\ldots\Delta k = \int_{-\infty}^{+\infty}\ldots\text{function}(k)\ldots \mathrm{d}k$$

$$f(x) = \frac{1}{2\pi} \sum_{k_n} \left[\int_{-\ell}^{\ell} f(y) e^{ik_n y}\, dy \right] e^{-ik_n x} \Delta k$$

$$\lim_{\ell \to \infty} \sum_{k_n} \ldots \text{function}(k_n) \ldots \Delta k = \int_{-\infty}^{+\infty} \ldots \text{function}(k) \ldots \mathrm{d}k$$

$$\hat{f}(k) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(y) e^{iky}\, dy \;=\; \mathcal{F}(f)(k) \quad \textbf{Fourier transform} \text{ of } f$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(k) e^{-ikx}\, dk \;=\; \mathcal{F}^{-1}(\hat{f})(x) \quad \hat{f} - \textbf{inverse Fourier transform} \text{ of } f$$

Now because of the special character of the this course we move from the spatial x, k notation to the temporal $t, \omega = 2\pi\nu$ notation

$$\hat{f}(\omega) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt \;=\; \mathcal{F}(f)(\omega) \quad \textbf{Fourier transform} \text{ of } f$$

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i\omega t}\, d\omega \;=\; \mathcal{F}^{-1}(\hat{f})(t) \quad \hat{f} - \textbf{inverse Fourier transform} \text{ of } f$$

Instead of the $\frac{1}{\sqrt{2\pi}}$ multipliers in the two definitions one can use any other as long as their product is $\frac{1}{2\pi}$ the factor necessary to retain the original function after subsequent transform + inverse transform.
It is also common to use the $\{1, \frac{1}{2\pi}\}$ pair:

$$\hat{f}(\omega) \equiv \int_{-\infty}^{\infty} f(y) e^{i\omega t}\, dt \;=\; \mathcal{F}(f)(\omega) \quad \textbf{Fourier transform} \text{ of } f$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i\omega t}\, d\omega \;=\; \mathcal{F}^{-1}(\hat{f})(t) \quad \hat{f} - \textbf{inverse Fourier transform} \text{ of } f$$

If $k \to 2\pi\nu$ and $x, y \to t$

$$\hat{f}(\nu) \equiv \int_{-\infty}^{\infty} f(t) e^{2\pi i\nu t}\, \mathrm{d}t$$

$$f(t) = \int_{-\infty}^{\infty} \hat{f}(\nu) e^{-2\pi i\nu t}\, \mathrm{d}\nu$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t') e^{i\omega(t'-t)} \, dt' \, d\omega$$

$$\int_{-\infty}^{\infty} e^{i\omega(t'-t)} \, d\omega = 2\pi\delta(t'-t)$$

$$\int_{-\infty}^{\infty} e^{2\pi i\nu t} \, \mathrm{d}\nu = \delta(t) \tag{1}$$

$$(f,g) = \int_{-\infty}^{\infty} f(t) \cdot \overline{g}(t) \, dt = \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i\omega t} \, d\omega \cdot \int_{-\infty}^{\infty} \overline{\hat{g}}(\omega') e^{i\omega' t} \, d\omega' \right) \, dt =$$

$$= \int_{-\infty}^{\infty} d\omega \; \hat{f}(\omega) \int_{-\infty}^{\infty} d\omega' \; \overline{\hat{g}}(\omega') \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \; e^{it(\omega'-\omega)} = \int_{-\infty}^{\infty} d\omega \; \hat{f}(\omega) \int_{-\infty}^{\infty} d\omega' \; \overline{\hat{g}}(\omega') \delta(\omega'-\omega) =$$

$$= \int_{-\infty}^{\infty} \hat{f}(\omega) \overline{\hat{g}}(\omega) \, d\omega = (\hat{f}, \overline{\hat{g}})$$

$$(f,g) = (\hat{f}, \hat{g})$$

1. $f(t)$ and $\hat{f}(\omega)$ equivalent. Each other's *dual*
2. $f(t)$ functions' space can be mapped to the space of the $\hat{f}(\omega)$ functions. They are each other's *dual* spaces.
3. $f(t)$ and $\hat{f}(\omega)$ are the different represenations of the same abstract f.

Exercises

Note, that above the definition with multipliers $1/\sqrt{2\pi}$ was used. Show that for the $\{1, 1/2\pi\}$ multipliers we have $(\hat{f}, \hat{g}) = 2\pi(f, g)$.

Properties of Fourier transforms:

1. **linearity**
$$\mathcal{F}(c_1 f + c_2 g)(\omega) = c_1 \hat{f}(\omega) + c_2 \hat{g}(\omega)$$
2. **time shift**
$$\mathcal{F}(f(t+b)) = e^{-i\omega b}\hat{f}(\omega)$$
3. **time/frequency scaling**
$$\mathcal{F}(f(at)) = \frac{1}{|a|}\hat{f}(\omega/a)$$
4. **frequency shift**
$$\mathcal{F}\left(f(t)e^{iat}\right) = \hat{f}\,(\omega + a)$$
5. **Parseval's theorem** (total power)
$$\int_{-\infty}^{\infty} |f(t)|^2 \mathrm{d}t = \int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 \mathrm{d}\omega$$

Exercises

Show the above properties of the Fourier transform.

Other:

1. $$\mathcal{F}\left(i^n t^n f(t)\right) = \frac{d^n}{d\omega^n}\hat{f}(\omega)$$

2. $$\mathcal{F}\left(\frac{d^n}{dt^n}f(t)\right) = (-i)^n \omega^n \hat{f}(\omega)$$

3. $$\mathcal{F}\left(\frac{1}{1+t^2}\right) = \sqrt{\frac{\pi}{2}}e^{-|\omega|}$$

4. $$\mathcal{F}\left(e^{-t^2/2}\right) = e^{-\omega^2/2}$$

5. $$\mathcal{F}\left(\cos\omega_0 t\right) = \frac{1}{2}\left[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)\right] \;, \qquad \mathcal{F}\left(\sin\omega_0 t\right) = \frac{1}{2i}\left[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)\right]$$

6. $$\mathcal{F}\left((a+it)^{-v}\right) = \begin{cases} \sqrt{2\pi}\,\omega^{v-1}e^{-a\omega}/\Gamma(v) & \omega > 0 \\ 0 & \omega < 0 \end{cases} \qquad (\{a, v\} > 0)$$

7. $$\mathcal{F}\left(\frac{e^{-\lambda t}}{a+e^{-t}}\right) \quad (a > 0,\ 0 < \lambda < 1) \quad = \sqrt{\frac{\pi}{2}}a^{\lambda-1-i\omega}\csc(\pi\lambda - i\pi\omega)$$

8. $$\mathcal{F}\left(\text{Ш}_T(t)\right) = T\,\text{Ш}_{1/T}(\nu)\ .$$

Proof

$$\mathcal{F}(\text{Ш}_T) = T\sum_n \int \delta(t-nT)e^{2\pi i\nu t}\mathrm{d}t = T\sum_n e^{2\pi i\nu nT} = \sum_n \delta\left(\nu - \frac{n}{T}\right)$$

from the Fourier series of the Dirac comb:
$$\sum_{n=-\infty}^{\infty} e^{in\frac{2\pi}{L}x} = L\cdot\sum_{n=-\infty}^{\infty}\delta(x-nL)$$

9. For a T periodic function $f(t)$ its Fourier-transform is an infinite set of equidistant Dirac-deltas at $1/T$ away from one another.

$$\mathcal{F}\left(f(t)\right) = \sum_{n=-\infty}^{+\infty} c_n \delta\left(f - \frac{n}{T}\right)\ ,$$

Proof

Use The Fourier-expansion of periodic functions and apply (1)